

Harmonic Function

$z = u + iv$
 ☐ Laplace's Equation: $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$
 ☐ Harmonic Function: (x, y)

$$u = x^2 - y^2 - 2x \quad \left| \begin{array}{l} \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2} = \\ \frac{\partial u}{\partial y}, \frac{\partial^2 u}{\partial y^2} = \end{array} \right.$$

☐ Determine which of the following functions are harmonic

i) $u = 2xy + 3xy^2 - 2y^3$

ii) $u = x e^x \cos y - y e^x \sin y$

iii) $u = e^{-x} (x \sin y - y \cos y)$

iv) $u = 2x - x^3 + 3xy^2$

v) $u = 2xy + y^3 - 3x^2y$

vi) $u = e^x \cos y$

vii) $u = -x^3 + 3xy^2 + 2x + 1$

Solution:

Let $y = c$

① $u = 2xy + 3xy^2 - 2y^3$

$$\therefore \frac{\partial u}{\partial x} = 2y \cdot 1 + 3y^2 \cdot 1 - 0 = 2y + 3y^2$$

$$\frac{\partial^2 u}{\partial x^2} = 0 + 0 = 0 \quad \text{--- ①}$$

$$\left[\begin{array}{l} \frac{d}{dx}(c) = 0 \\ \frac{d}{dx}(cx) = c \frac{d}{dx}(x) = c \cdot 1 \end{array} \right.$$

$$\frac{\partial u}{\partial x^2} = 0 + 0 = 0 \quad \text{--- (1)}$$

Again,
Let, $x=c$

$$u = 2xy + 3xy^2 - 2y^3$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= 2x \cdot 1 + 3x \cdot 2y - 2 \cdot 3y^2 \\ &= 2x + 6xy - 6y^2 \end{aligned}$$

$$\frac{\partial^2 u}{\partial y^2} = 0 + 6x \cdot 1 - 6 \cdot 2y = 6x - 12y \quad \text{--- (2)}$$

$$\begin{aligned} \text{(1) + (2)} \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0 + 6x - 12y \\ &= 6x - 12 \neq 0 \end{aligned}$$

Since, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \neq 0$

So, u is not a harmonic function.

$$\text{(3)} \quad u = e^{-x}(x \sin y - y \cos y)$$

$$u = \frac{e^{-x}}{u} x \sin y - e^{-x} y \cos y \quad \left[\frac{d}{dx}(e^{-x}) = -e^{-x} \right]$$

$$\therefore \frac{\partial u}{\partial x} = \sin y \left[e^{-x} \frac{\partial}{\partial x}(x) + x \frac{\partial}{\partial x}(e^{-x}) \right] + e^{-x} y \cos y$$

$$\begin{aligned} &= \sin y [e^{-x} \cdot 1 - x e^{-x}] + e^{-x} y \cos y \\ e^{-x} &= \sin y - \frac{x e^{-x} \sin y}{e^{-x}} + e^{-x} y \cos y \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= -e^{-x} \sin y - [e^{-x} - x e^{-x}] \sin y - e^{-x} y \cos y \\ &= -e^{-x} \sin y - e^{-x} \sin y + x e^{-x} \sin y - e^{-x} y \cos y \\ &= -2e^{-x} \sin y + x e^{-x} \sin y - e^{-x} y \cos y \end{aligned}$$

$$= 2e^{-x} \sin y + x e^{-x} \sin y - e^{-x} \cos y \quad \text{--- (1)}$$

Again,

$$u = e^{-x} x \sin y - e^{-x} y \cos y$$

$$\therefore \frac{\partial u}{\partial y} = e^{-x} x \cos y - e^{-x} \left[y \frac{\partial}{\partial y} (\cos y) + \cos y \frac{\partial}{\partial y} (y) \right]$$

$$= e^{-x} x \cos y - e^{-x} [-y \sin y + \cos y]$$

$$= e^{-x} x \cos y + e^{-x} y \sin y - e^{-x} \cos y$$

$$\therefore \frac{\partial^2 u}{\partial y^2} = -e^{-x} x \sin y + e^{-x} \left[y \frac{\partial}{\partial y} (\sin y) + \sin y \frac{\partial}{\partial y} (y) \right] + e^{-x} \sin y$$

$$= -e^{-x} x \sin y + e^{-x} (y \cos y + \sin y) + e^{-x} \sin y$$

$$= -e^{-x} x \sin y + e^{-x} y \cos y + e^{-x} \sin y + e^{-x} \sin y$$

$$= 2e^{-x} \sin y - e^{-x} x \sin y + e^{-x} y \cos y \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$\therefore$, u is harmonic function.